Problem Set 3
Due: Mar. 15, 2021

1. Section 1.5 Problem 31.

Communication through a noisy channel. A source transmits a message (a string of symbols) through a noisy communication channel. Each symbol is 0 or 1 with probability p and $1-p$, respectively, and is received incorrectly with probability ϵ_0 and ϵ_1, respectively (see Fig. 1.18). Errors in different symbol transmissions are independent.

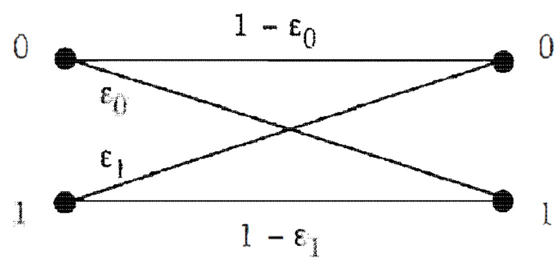

Figure 1.18: Error probabilities in a binary communication channel.

(a) What is the probability that the kth symbol is received correctly?

(b) What is the probability that the string of symbols 1011 is received correctly?

(c) In an effort to improve reliability, each symbol is transmitted three times and the received string is decoded by majority rule. In other words, a 0 (or 1) is transmitted as 000 (or 111, respectively), and it is decoded at the receiver as a 0 (or 1) if and only if the received three-symbol string contains at least two 0s (or 1s, respectively). What is the probability that a 0 is correctly decoded?

(d) For what values of ϵ_0 is there an improvement in the probability of correct decoding of a 0 when the scheme of part (c) is used?

(e) Suppose that the scheme of part (c) is used. What is the probability that a symbol was 0 given that the received string is 101?

Solution:

(a) Let A be the event that a 0 is transmitted. Using the total probability theorem, the desired probability is

$$P(A)(1-\epsilon_0) + (1-P(A))(1-\epsilon_1) = p(1-\epsilon_0) + (1-p)(1-\epsilon_1).$$

(b) By independence, the probability that the string 1011 is received correctly is

$$(1-\epsilon_0)(1-\epsilon_1)^3.$$

(c) In order for a 0 to be decoded correctly, the received string must be 000, 001, 010, or 100. Given that the string transmitted was 000, the probability of receiving 000 is $(1-\epsilon_0)^3$, and the probability of each of the strings 001, 010, and 100 is $\epsilon_0(1-\epsilon_0)^2$. Thus, the probability of correct decoding is

$$3\epsilon_0(1-\epsilon_0)^2 + (1-\epsilon_0)^3.$$

(d) When the symbol is 0, the probabilities of correct decoding with and without the scheme of part (c) are $3\epsilon_0(1-\epsilon_0)^2+(1-\epsilon_0)^3$ and $1-\epsilon_0$, respectively. Thus, the probability is improved with the scheme of part (c) if

$$3\epsilon_0(1-\epsilon_0)^2 + (1-\epsilon_0)^3 > (1-\epsilon_0),$$

or

$$(1-\epsilon_0)(1+2\epsilon_0) > 1,$$

which is equivalent to $0 < \epsilon_0 < 1/2$.

(e) Using Bayes' rule, we have

$$P(0|101) = \frac{P(0)P(101|0)}{P(0)P(101|0) + P(1)P(101|1)}.$$

The probabilities needed in the above formula are

$$P(0) = p, \qquad P(1) = 1-p, \qquad P(101|0) = \epsilon_0^2(1-\epsilon_0), \qquad P(101|1) = \epsilon_1(1-\epsilon_1)^2.$$

2. Section 1.5 Problem 38.

The problem of points. Telis and Wendy play a round of golf (18 holes) for a $10 stake, and their probabilities of winning on any one hole are p and $1-p$, respectively, independent of their results in other holes. At the end of 10 holes, with the score 4 to 6 in favor of Wendy, Telis receives an urgent call and has to report back to work. They decide to split the stake in proportion to their probabilities of winning had they completed the round, as follows. If p_T and p_W are the conditional probabilities that Telis and Wendy, respectively, are ahead in the score after 18 holes given the 4-6 score after 10 holes, then Telis should get a fraction $p_T/(p_T+p_W)$ of the stake, and Wendy should get the remaining $p_W/(p_T+p_W)$. How much money should Telis get? *Note:* This is an example of the, so-called, problem of points, which played an important historical role in the development of probability theory. The problem was posed by Chevalier de Méré in the 17th century to Pascal, who introduced the idea that the stake of an interrupted game should be divided in proportion to the players' conditional probabilities of winning given the state of the game at the time of interruption. Pascal worked out some special cases and through a correspondence with Fermat, stimulated much thinking and several probability-related investigations.

Solution: We have

$$p_T = P(\text{at least 6 out of 8 remaining holes are won by Telis}),$$
$$p_W = P(\text{at least 4 out of 8 remaining holes are won by Wendy}).$$

Using the binomial formulas,

$$p_T = \sum_{k=6}^{8} \binom{8}{k} p^k (1-p)^{8-k}, \qquad p_W = \sum_{k=4}^{8} \binom{8}{k} (1-p)^k p^{8-k}.$$

The amount of money that Telis should get is $10p_T/(p_T + p_W)$ dollars.

3. Show that the conditional independence of A and B given C neither implies, nor is implied by, the independence of A and B. For which events C is it the case that. for all A and B, the events A and B are independent if and only if they are conditionally independent given C?

   Solution:

   (a) Flip two coins; let A be the event that the first shows H, let B be the event that the second shows H, and let C be the event that they show the same. Then A and B are independent, but not conditionally independent given C.

   (b) Roll two dice; let A be the event that te smaller is 3, let B be the event that the larger is 6, and let C be the event that the smaller score is no more than 3, and the larger is 4 or more. Then A and B are conditionally independent given C, but not independent.

   (c) The definitions are equivalent if $P(C) = 1$.

4. Jane has three children, each of which is equally likely to be a boy or a girl independently of the others. Define the events:

$$A = \{\text{all the children are of the same sex}\},$$
$$B = \{\text{there is at most one boy}\},$$
$$C = \{\text{the family includes a boy and a girl}\}.$$

   (a) Show that A is independent of B, and that B is independent od C.

   (b) Is A independent of C?

   (c) Do these results hold if boys and girls are not equally likely?

   (d) Do these results hold if Jane has four children?

   Solution:

   (a) $P(A \cap B) = \frac{1}{8} = \frac{1}{4} \cdot \frac{1}{2} = P(A)P(B)$, and $P(B \cap C) = \frac{3}{8} = \frac{1}{2} \cdot \frac{3}{4} = P(B)P(C)$.

(b) $P(A \cap C) = 0 \neq P(A)P(C)$.

(c) Only in the trivial cases when children are either almost surely boys or almost surely girls.

(d) No.